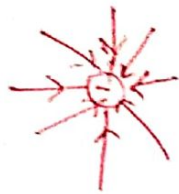
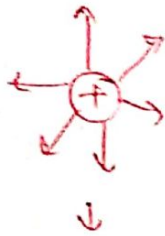


Electric lines of force :

- Not intersect each other
- Always normal to surface of body
- ↗ ↘ repel
- ↗ ↙ attract
- Not exists inside the conductor



Electric field due to charge q at a distance r

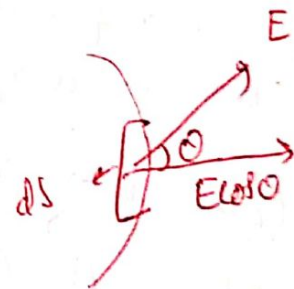
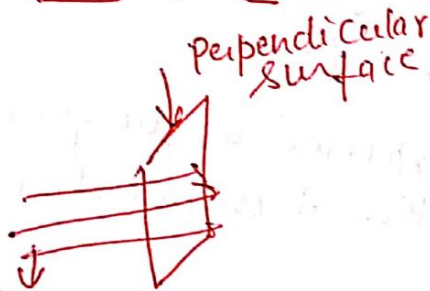
$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \quad \left(\text{N/C or lines/m}^2 \right)$$

→ Sphere of radius r has a surface area is given by

$$S = 4\pi r^2$$

$$E \times 4\pi r^2 = \frac{q}{\epsilon_0}$$

Flux of Electric field : →



no. of lines of force passing through small area

→ $d\phi$ = Electric flux = $E \cdot dS = E dS \cos \theta$

$$\Rightarrow \phi = \oint_S E \cdot dS$$

→ The number of lines of force passing through a unit area, which is perpendicular to the direction of lines of force.

Electric flux density

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \quad \hat{r} \rightarrow \text{unit vector in radial dir}^n$$

$$\frac{q}{4\pi r^2} \hat{r} = \boxed{\epsilon_0 E = D}$$

units $\rightarrow$ Coloumb/m²

also known as electric flux density
or electric displacement density
or electric inductance

Surface charge density

Gauss's Theorem or Gauss law of Electrostatics :-

total outward flux of an electric field over a closed surface is equal to $\frac{1}{\epsilon_0}$ times the total net charge contained in a volume enclosed by the surface

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \frac{q}{\epsilon_0} \rightarrow \text{permittivity of free space}$$

Electric field intensity

$$\phi = \oint \mathbf{E} \cdot d\mathbf{s} = \frac{q}{\epsilon_0}$$

$$= \int \epsilon_0 \mathbf{E} \cdot d\mathbf{s} = q$$

$$\boxed{\oint \mathbf{D} \cdot d\mathbf{s} = q}$$

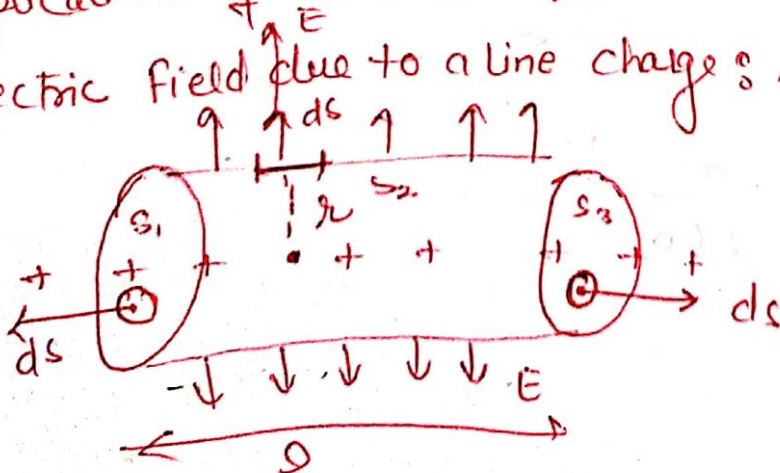
$$K = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

$$\Rightarrow \epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N m}^2}$$

or Farad/metre

Applications of Gauss law :-

1) Electric field due to a line charge :-



$$\int_S \vec{E} \cdot d\vec{s} = \int_{S_1} \vec{E} \cdot d\vec{s} + \int_{S_2} \vec{E} \cdot d\vec{s} + \int_{S_3} \vec{E} \cdot d\vec{s}$$

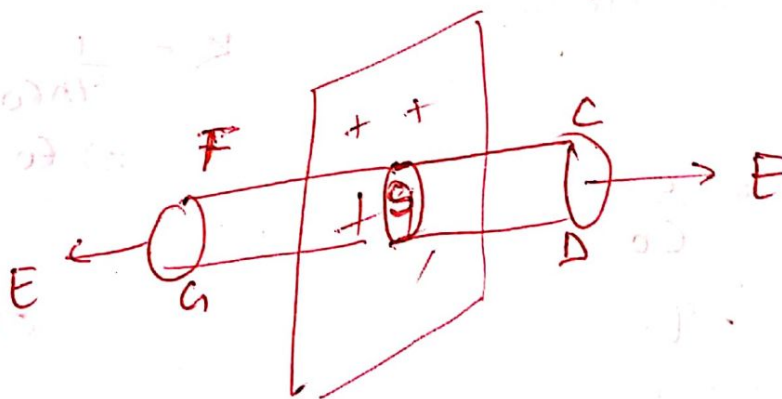
$$\Rightarrow \int_{S_2} \vec{E} \cdot d\vec{s} = E \int_{S_2} ds = E 2\pi r l$$

$$\boxed{\rho = \frac{q}{l}} \Rightarrow \rho l = q$$

$$2\pi r l E = \frac{\rho l}{\epsilon_0}$$

$$E = \frac{\rho}{2\pi \epsilon_0 r}$$

⑥ Electric field due to sheet of charge



$$\int_{CD} \vec{E} \cdot d\vec{s} + \int_{FG} \vec{E} \cdot d\vec{s} = \frac{\sigma S}{\epsilon_0}$$

$$E \cdot S + E \cdot S = \frac{\sigma S}{\epsilon_0}$$

$$\boxed{E = \frac{\sigma}{2\epsilon_0}}$$

Electric field due to charge on a conductor



Since the electric field within the conductor under electrostatic condition is zero, so electric flux is zero.

(d) Electric field due to a charged Non-conducting sphere :-

(i) within the sphere

$$\rho_v = \frac{q}{\frac{4}{3}\pi a^3}$$

$$\Phi_E = \int_S \mathbf{E} \cdot d\mathbf{s} = E_1 ds \cos 0^\circ = \int E_1 ds$$

$$= E_1 (4\pi r^2)$$

for radius r , charge enclosed

$$\rho_v \cdot \frac{4}{3}\pi r^3 = \frac{1}{\frac{4}{3}\pi a^3} \cdot \frac{4}{3}\pi r^3 = \frac{r^3}{a^3}$$

Accⁿ to Gauss law

$$E_1 \cdot 4\pi r^2 = \frac{q}{\epsilon_0} \frac{r^3}{a^3}$$

$$E_1 = \frac{qr}{4\pi\epsilon_0 a^3}$$

(ii) on the surface of the sphere

$$\Phi = \int_S \mathbf{E}_2 \cdot d\mathbf{s} = E_2 \cdot 4\pi a^2$$

entire charge (q)

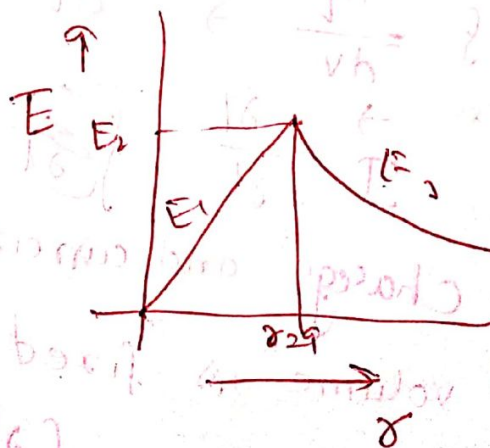
$$E_2 \cdot 4\pi a^2 = \frac{q}{\epsilon_0}$$

$$\Rightarrow E_2 = \frac{q}{4\pi\epsilon_0 a^2}$$

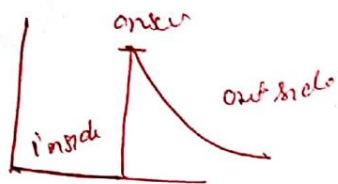
(iii) outside the sphere

$$E_3 \cdot 4\pi r^2 = \frac{q}{\epsilon_0}$$

$$E_3 = \frac{q}{4\pi\epsilon_0 r^2}$$



e) Electric field around a Non conducting hollow sphere



Limitation of Gauss law :-

→ Gauss law depends on symmetry, if we can't show that symmetry exists then we can't use Gauss law.

→ have to look that in which coordinate of field varies

→ Coulomb law → $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$

$$\frac{F_{12}}{q_1} = \left(\frac{1}{4\pi\epsilon_0} \frac{q_2}{r^2} \right) = E$$

$$\boxed{F_{12} = E q_1}$$

Continuity Equations :-
It represents conservation of charge (charge can neither be created nor destroyed).
incoming charge = outgoing charge

It relates current density to charge density.

$$J = \frac{dq}{ds \cdot dt} \Rightarrow dI = J \cdot ds$$

$$I = \int J \cdot ds = \text{using Gauss's closed surface} \int \nabla \cdot J \, dV \rightarrow \text{for fixed volume}$$

$$\rho = \frac{dq}{dv} \Rightarrow dq = \rho \cdot dv$$

$$\Rightarrow I = \frac{dq}{dt} = \frac{d}{dt} \int \rho \, dv$$

Charge and current are in opposite dirⁿ

→ When volume is fixed

$$\text{then } I = - \int \frac{\partial \rho}{\partial t} \, dv$$

Comparison

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}$$

$$\Rightarrow \boxed{\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0}$$

If there is no source and sink



$$\boxed{\nabla \cdot \mathbf{J} = 0}$$

Lorentz Force

$$\mathbf{F}_m = q(\mathbf{v} \times \mathbf{B}) = qvB \sin \theta$$

Electromagnetic force

$$\mathbf{F} = \mathbf{F}_e + \mathbf{F}_m = q\mathbf{E} + q(\mathbf{v} \times \mathbf{B})$$

Magnetic flux

$$\Phi_B = \int \vec{B} \cdot d\vec{s}$$

$$d\Phi = \vec{B} \cdot \hat{n} ds = B ds \cos \theta$$



$$\oint \vec{B} \cdot d\vec{s} = \int \nabla \cdot \mathbf{B} dV = 0$$



magnetic loop
no source & sink

$$\Rightarrow \nabla \cdot \mathbf{B} = 0$$

$\Rightarrow$ magnetic monopoles can't exist in nature
only (N) or (S)

Faraday's Law :-

$$e = \text{emf} = -\frac{d\Phi}{dt}$$

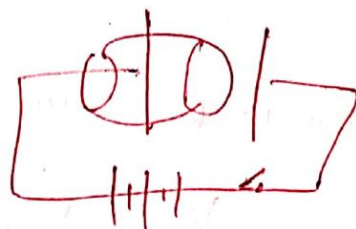
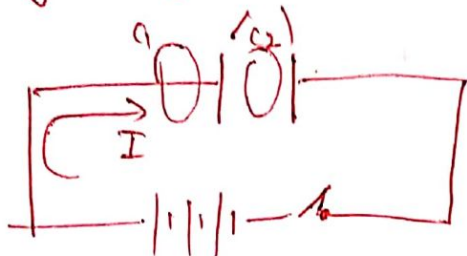


magnetic flux

change in magnetic field
induce electric field (emf)

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

$\vec{B}$ along any closed loop C is proportional to current I passing through closed loop.



Ampere

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I$$

but

$$\oint_C \vec{B} \cdot d\vec{l} = 0$$

If C_1 & C_2 are close then

$$\oint_{C_1} \vec{B} \cdot d\vec{l} = \oint_{C_2} \vec{B} \cdot d\vec{l}$$

Both result are not same
So there must be modification
required in Ampere's law

Displacement Current :-

$$E \cdot S = \frac{q}{\epsilon_0}$$

$$E \cdot S =$$

It comes into existence, In ~~addition~~ addition to Conduction current, whenever the electric field and hence the electric flux changes with time.

$$I_d = \epsilon_0 \frac{d\phi_E}{dt}$$

$\rightarrow \phi_E = \underline{E \cdot S} \rightarrow$ electric flux

By using Faraday's law symmetry change in electric field must induce magnetic field.

so total current

$$I_c + I_d = I_c + \epsilon_0 \frac{d\phi_E}{dt}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left[I_c + \epsilon_0 \frac{d\phi_E}{dt} \right]$$

modified Ampere's law.

$$\begin{aligned} \oint \vec{B} \cdot d\vec{l} &= \mu_0 \epsilon_0 \frac{d\phi_E}{dt} \\ &= \mu_0 \epsilon_0 \frac{1}{\epsilon_0} \frac{dq}{dt} \\ &= \mu_0 I \end{aligned}$$

$$\begin{aligned} \oint \vec{E} \cdot d\vec{s} &= \frac{q}{\epsilon_0} \\ \phi_E & \end{aligned}$$

Properties of Displacement Current

→ Displacement current exists whenever there is a change in electric flux. Unlike conduction current, it doesn't exist under steady condition.

→ It is not a current. It only adds to current density in Ampere's circuital law. As it produces magnetic field, so it is called a current.

→ Together with conduction current, it satisfies the property of continuity.

→ magnitude of it is equal to rate of displacement of charge from one capacitor plate to other.

$$I_c = \sigma E, \quad I = \frac{V}{R}$$

$$I_d = \frac{dQ}{dt} = \epsilon_0 \frac{dE}{dt}$$

But for constant E
 $I_d = 0$

for constant E
 $T \neq 0$

Maxwell Equations: -

1. Gauss law for electric field:-

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \text{or} \quad \oint_S \vec{E} \cdot d\vec{s} = \frac{1}{\epsilon_0} \int_V \rho dV$$

2. Gauss law for magnetic field:-

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \oint_S \vec{B} \cdot d\vec{s} = 0$$

3. Faraday's law of Induction:-

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$
$$\oint_C \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t} \int_S \vec{B} \cdot d\vec{s} = -\frac{\partial \Phi_B}{\partial t}$$

4. Modified Ampere's Law

$$\vec{\nabla} \times \vec{B} = \mu_0 \left[\vec{J} + \frac{\partial \vec{D}}{\partial t} \right] = \mu_0 \left[\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right]$$

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 \int_S \left[\vec{J} + \frac{\partial \vec{D}}{\partial t} \right] \cdot d\vec{s}$$

$$= \mu_0 \int_S \left[\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right] \cdot d\vec{s}$$

$$= \mu_0 \int_S \vec{J} \cdot d\vec{s} + \mu_0 \epsilon_0 \int_S \frac{\partial \vec{E}}{\partial t} \cdot d\vec{s}$$

$$= \mu_0 I + \mu_0 \epsilon_0 \frac{\partial \Phi_E}{\partial t}$$